

Article

Research on Speed Control of Permanent Magnet Synchronous Motor Based on Improved Fast Terminal Sliding Mode with Adaptive Control Law

Mingyuan Hu ^{1,*} , Lei Zhang ² , Ran Tao ³ and Ping Wang ¹
¹ School of Electrical Engineering, Shandong Huayu University of Technology, Dezhou 253000, China; wp@huayu.edu.cn

² School of Electrical Engineering, Beijing Jiaotong University, Beijing 100091, China; 22110464@bjtu.edu.cn

³ School of Electrical Engineering, Southeast University, Nanjing 210096, China; cumt_ran@163.com

* Correspondence: hmy@huayu.edu.cn; Tel.: +86-0534-7023793

Abstract

Aiming at the control performance degradation of permanent magnet synchronous motor (PMSM) drive systems caused by uncertainties of internal and external disturbances, a robust control algorithm integrating an improved fast terminal sliding mode (IFTSM) surface with a novel adaptive reaching law (NARL) is proposed. A dynamic model of PMSM with disturbances is established, and an improved fast terminal sliding mode surface is designed. By introducing nonlinear terms and error derivative feedback mechanisms, the finite-time rapid convergence of system states is achieved, while solving the singularity problem of traditional terminal sliding mode control. Combined with the novel adaptive reaching law strategy, a state-dependent gain adjustment function is used to dynamically optimize the balance between reaching speed and chattering, enhancing the smoothness of the system's dynamic response. Through the synergy of the finite-time convergence characteristic of the improved sliding mode surface and the novel adaptive reaching law, the proposed algorithm significantly enhances the system's anti-interference capability against load mutations and parameter time variations. Experiment results demonstrate that under complex working conditions, the algorithm achieves superior speed tracking accuracy and current stability, providing a control solution with strong anti-interference capability and fast response for PMSM speed control systems.

Keywords: PMSM; ITFSMC; NARL; speed control



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1. Introduction

Permanent Magnet Synchronous Motors (PMSMs) are widely used due to their small size, simple structure, and fast dynamic response. However, in practical operation, unknown dynamic friction, system uncertainties, and external disturbances affect system performance, causing inconveniences in actual production, normal operation of electric vehicles, etc. Traditional PI control features simple methodology and fast response but performs poorly in the face of unknown external disturbances and internal parameter perturbations [1]. Therefore, scholars have proposed adaptive control [2], robust control [3], fuzzy control [4], model predictive control [5], etc. Sliding Mode Control (SMC) is widely applied in PMSM control due to its good transient performance and strong robustness [6].

Traditional SMC cannot guarantee finite-time convergence, while Fast Terminal Sliding Mode Control (FTSMC) ensures finite-time convergence of the system [7]. However,

since FTSMC introduces nonlinear terms in the approaching stage, the control quantity may grow unboundedly when the system state approaches the equilibrium point, leading to singular point problems [8]. References have designed a sliding surface with state-dependent variable exponential coefficients to avoid the singular phenomenon in terminal sliding mode control and achieve fixed-time convergence of the PMSM drive system [9]. Furthermore, in the field of multi-agent cooperative control (e.g., islanded microgrids), researchers have achieved convergence performance independent of initial states by designing a fully distributed fixed-time control strategy with dual-power nonlinear terms and adaptive gain adjustment [10], and this method provides a reference for optimizing fixed-time sliding mode control of PMSM. Reference [11] puts forward a continuous fast TSMC, which shows strong robustness to external disturbances and features fast response performance. In [12], building on TSMC, develops a fast TSMC with a linear variable term incorporated, which is applied to the motion control of robotic manipulators. Nevertheless, the presence of the sign function causes chattering in the system. In [13], the arctangent function replaces the sign function, effectively reducing chattering. The integral non-singular fast terminal sliding surface designed in reference [14] combines an adaptive function, using the smoothing property of integral operation to avoid the singular phenomenon caused by the denominator being zero when the error approaches zero in the traditional terminal sliding surface. However, its first-order system structure inherently cannot completely eliminate the chattering caused by high-frequency switching.

Chattering in SMC affects system control accuracy. Thus, scholars have proposed various chattering suppression methods, including reaching laws [15–17], boundary layer methods [18], and disturbance observer methods [19]. The chattering problem in SMC arises from the high-frequency switching of the sliding surface due to insufficient convergence of the system trajectory under the influence of the reaching law. Traditional reaching laws often slow down the system convergence rate while suppressing chattering. To address this, ref. [20] constructs a hierarchical gain adjustment mechanism: high-gain power terms are used for fast approaching in the early stage of the approaching phase, and low-gain continuous functions are switched when the system state enters a preset neighborhood. On this basis, reference [21] proposes a time-varying gain adaptive framework, whose core innovation lies in designing a gain adjustment function related to the state. This function dynamically adjusts the switching term coefficient based on the exponential decay characteristics of the tracking error to achieve collaborative optimization of the approaching speed and chattering intensity. The novel reaching law designed in reference [22] introduces a composite function containing all system state variables on the basis of the traditional exponential reaching law to suppress SMC chattering and improve the approaching speed, and combines an inverse calculation anti-saturation method to avoid integral saturation. However, its calculation is too complex, and parameter tuning is difficult.

Traditional SMCs have obvious limitations in PMSM control: conventional linear sliding modes struggle to achieve finite-time convergence with slow dynamic responses; traditional terminal sliding modes are prone to singularity issues, leading to unbounded control quantities; meanwhile, they suffer from insufficient steady-state accuracy, significant chattering, and weak robustness against sudden load changes and parameter perturbations. To address the above issues, this paper proposes an improved fast terminal sliding mode surface design based on a novel adaptive control law. A nonlinear term and an integral feedback mechanism are incorporated into the sliding mode surface to eliminate steady-state errors. In addition, compared with traditional control laws, the novel adaptive control law proposed in this paper utilizes a state-dependent gain adjustment function and a nonlinear damping term to enhance system robustness while suppressing sliding mode chattering. A Lyapunov function is employed to conduct rigorous analysis on system

stability. Moreover, simulation studies are carried out via the MATLAB/Simulink platform, and the results are compared with those of the SMC-TRL, SMC-[7], and ITFSMC-TRL methods to verify the effectiveness of the proposed control strategy in terms of dynamic response, disturbance rejection capability, and steady-state accuracy.

2. System Description

This chapter introduces the mathematical model in the dq reference frame. To derive the model, the following assumptions are made:

- The spatial magnetic field is sinusoidally distributed;
- Stator core saturation is neglected;
- Iron losses are neglected.

Motor parameter perturbations are considered in the subsequent disturbance model (Equation (2)). Based on the above assumptions, the mathematical equations of the PMSM are expressed as follows [18,23–25].

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \\ u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \psi_f) \\ \frac{d\omega_e}{dt} = \frac{n_p}{J} (T_e - T_L - B\omega_m) \\ T_e = \frac{3}{2} n_p \psi_f i_q \end{cases}, \quad (1)$$

where u_d denotes the d-axis voltage component of the stator, and u_q represents the q-axis voltage component of the stator. i_d stands for the d-axis current component of the stator, while i_q indicates the q-axis current component of the stator. L_d refers to the d-axis inductance of the stator winding, with L_q corresponding to the q-axis inductance of the stator winding. ω_e is the electrical angular velocity; B denotes the damping coefficient; and T_e represents the electromagnetic torque. J stands for the moment of inertia, T_L indicates the load torque, and ω_m is the mechanical angular velocity. ψ_f refers to the permanent magnet flux linkage, R_s denotes the stator resistance, and n_p represents the number of pole pairs.

When the motor undergoes parameter perturbations, including fluctuations in stator resistance, shifts in stator inductance, and permanent magnet demagnetization faults, Equation (1) ceases to be valid; instead, the mathematical formulations of the PMSM can be expressed as [26]

$$\begin{cases} u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q + \Delta u_d \\ u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e (L_d i_d + \psi_f) + \Delta u_q \\ \frac{d\omega_e}{dt} = \frac{n_p}{J} (T_e - T_L - B\omega_m) + \Delta P_n \\ T_e = \frac{3}{2} n_p \psi_f i_q + \Delta T_e \end{cases}, \quad (2)$$

where Δu_d represents the d-axis voltage disturbance in complex operating scenarios, while Δu_q denotes the q-axis voltage disturbance under complex operational conditions. ΔT_e stands for the electromagnetic torque disturbance in complex operating environments, and ΔP_n refers to the disturbance arising from changes in the moment of inertia J and damping

coefficient B . Through the combination of Equations (1) and (2), the subsequent result can be derived [27]:

$$\frac{d\omega_e}{dt} = \begin{bmatrix} \frac{3n_p^2}{2f} \psi_f i_q - \frac{B}{f} \omega_e + \\ \frac{n_p}{f} (\Delta T_e - T_L + \Delta T_L) + \Delta P_n \end{bmatrix}, \quad (3)$$

3. Design Speed Loop IFITSMC Controller

3.1. Design of IFITSM Surface

Define the state error

$$e = \omega_e^* - \omega_e, \quad (4)$$

where ω_e^* is the given rotational speed. To achieve finite-time fast convergence of the system states, the sliding surface is designed as

$$s = c_1 e + \dot{e} + \alpha |e|^{\alpha_1} \text{sgn}(e) + c_2 \int e dt, \quad (5)$$

where $0 < \alpha_1 < 1$, which enhances the control effect under small errors and helps improve the steady-state accuracy. $c_1 > 0$, $\alpha > 0$, $c_2 > 0$.

In the sliding surface, the proportional term $c_1 e$ and derivative term $\dot{e}$ provide system stiffness and damping, respectively, dominating the dynamic response process to ensure rapid error convergence and stability. The nonlinear term $\alpha |e|^{\alpha_1} \text{sgn}(e)$ leverages the strong gain characteristic of $|e|^{\alpha_1}$ relative to the linear term under small errors ($|e| \rightarrow 0$), enhancing the convergence capability for tiny errors and improving steady-state accuracy. The integral term $c_2 \int e dt$ accumulates errors, eliminates steady-state errors, and enhances the steady-state performance of the system.

Theorem 1. For the sliding surface $s = 0$, the error convergence time T can be expressed as

$$T = \frac{1}{(1 - \alpha_1)c_1} \ln \left(\frac{c_1 |e(0)|^{1-\alpha_1} + \alpha}{\alpha} \right), \quad (6)$$

where $|e(0)|$ is the initial absolute error.

Proof of Theorem 1. In the transient analysis, the derivative of the integral term $c_2 \int e dt$, has a negligible impact on the transient process. Therefore, the sliding surface is simplified as:

$$\dot{e} = -c_1 e - \alpha |e|^{\alpha_1} \text{sgn}(e), \quad (7)$$

When $e > 0$, at this time $\text{sgn}(e) = 1$, and the equation is

$$\dot{e} = -(c_1 e + \alpha e^{\alpha_1}), \quad (8)$$

Separate the variables and integrate

$$\int_{e(0)}^0 \frac{de}{c_1 e + \alpha e^{\alpha_1}} = \int_0^T dt, \quad (9)$$

Let $u = e^{1-\alpha_1}$, then $du = (1 - \alpha_1)e^{-\alpha_1} de$. Substituting this into the equation yields

$$T_+ = \frac{1}{(1 - \alpha_1)c_1} \ln \left(\frac{c_1 e(0)^{1-\alpha_1} + \alpha}{\alpha} \right), \quad (10)$$

When $e < 0$, let $v = -e > 0$, the equation is

$$\dot{v} = c_1 v + \alpha v^{\alpha_1}, \quad (11)$$

Similarly, integrating gives

$$T_- = \frac{1}{(1 - \alpha_1)c_1} \ln \left(\frac{c_1 |e(0)|^{1-\alpha_1} + \alpha}{\alpha} \right), \quad (12)$$

Due to the symmetry of the error sign, $T_+ = T_-$, so the convergence time is given by Equation (6).

The proof of Theorem 1 is completed. $\square$

By designing the reaching law, the dynamic performance of the system is improved. The reaching law designed by applying TRL (Terminal Reaching Law) can be expressed as

$$\dot{s} = -k_1 \text{sign}(s) - k_2 s, \quad (13)$$

where k_1, k_2 are positive gain coefficients, the derivative of Equation (5) is expressed as

$$\dot{s} = \ddot{e} + (c_1 + \alpha \alpha_1 |e|^{\alpha_1-1}) \dot{e} + c_2 e, \quad (14)$$

Combining Equations (3), (13) and (14),

$$i_q^* = \frac{J}{1.5p\psi_f} \left[(c_1 + \alpha \alpha_1 |e|^{\alpha_1-1}) \dot{e} + c_2 e - \frac{B}{J} \omega_m + k_1 \text{sign}(s) + k_2 s \right], \quad (15)$$

Design the IFSM surface as shown in Figure 1.

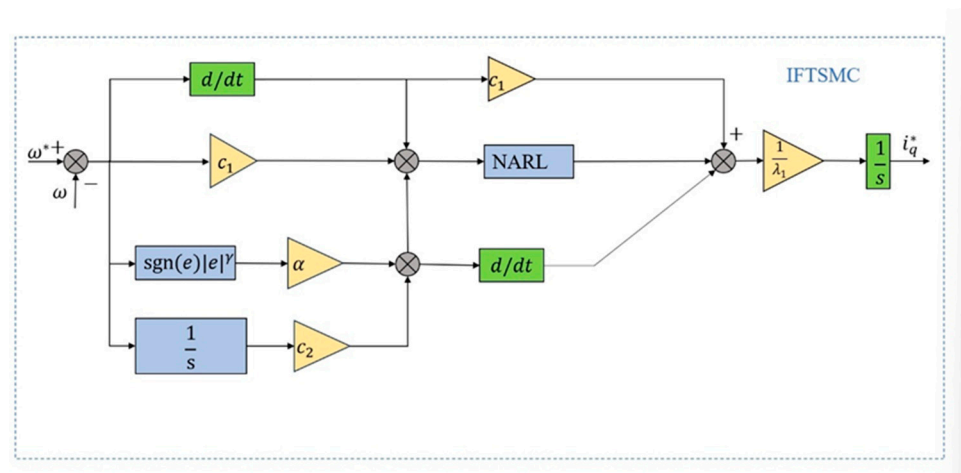


Figure 1. Control Block Diagram of IFSM Surface.

3.2. NARL Design

In the traditional reaching law, the linear term increases the system convergence speed, but it also introduces a large control law that causes system overshoot and affects control accuracy. Therefore, this paper proposes a novel adaptive law, which adjusts the magnitude of the control input according to the system error to improve control accuracy. The NARL is designed as

$$\begin{cases} \dot{s} = -f(e, s) \text{sgn}(s) - \beta_1 s - \beta_2 |s|^{0.5} \text{sgn}(s) \\ f(e, s) = \frac{\varepsilon_1 |e|}{\varepsilon_2 + (1 - \varepsilon_2) e^{-\delta |s|}} \dot{s} = -f(e, s) \text{sgn}(s) - \beta_1 s - \beta_2 |s|^{0.5} \text{sgn}(s) \\ \lim_{t \rightarrow 0} |e| = 0, \varepsilon_1 > 0, \delta > 0, \beta_1 > 0, \beta_2 > 0, 0 < \varepsilon_2 < 1 \end{cases}, \quad (16)$$

In Equation (16), the nonlinear switching term $-f(e, s)\text{sgn}(s)$ utilizes an adaptive gain positively correlated with the error $|e|$. In the large-error phase, it dominates the control through a strong switching gain to accelerate the transient convergence of the system; in the small-error phase, $f(e, s)$ decays with $|e|$, effectively avoiding the chattering problem of traditional sliding mode control and ensuring steady-state accuracy. The linear term $-\beta_1 s$ provides basic stability support, collaborating with the decayed nonlinear switching term in the small-error domain to dominate the smooth convergence of the sliding surface s , guaranteeing the system's steady-state control capability. The nonlinear damping term $-\beta_2 |s|^{0.5}\text{sgn}(s)$ further enhances the control robustness: when the sliding surface fluctuates significantly, $|s|^{0.5}$ provides strong damping to suppress drastic system changes; when the sliding surface approaches zero, the damping effect weakens, forming a nonlinear adjustment mechanism to improve the smoothness of the convergence process. The denominator of the adaptive gain $f(e, s)$ approaches ε_2 as $|s|$ increases, achieving dual adaptive adjustment of the error and the sliding surface state.

Theorem 2. *The reaching time of the NARL proposed in this paper is as follows:*

$$T_{NARL} = \frac{2}{\beta_1} \ln \left(\frac{\beta_1 \sqrt{|s(0)|} + \beta_2}{\beta_2} \right), \quad (17)$$

where when $|s| \rightarrow 0$, $f(e, s) \rightarrow 0$.

Proof of Theorem 2. When $s > 0$ (where $\text{sgn}(s) = 1$), the equation is simplified as:

$$\dot{s} = -f(e, s) - \beta_1 s - \beta_2 \sqrt{s}, \quad (18)$$

When $s \rightarrow 0$, $|e| \rightarrow 0$, thus $f(e, s) \rightarrow 0$, and the equation becomes: $\dot{s} = -\beta_1 s - \beta_2 \sqrt{s}$. Let $u = \sqrt{s}$, then

$$\dot{u} = \frac{\dot{s}}{2\sqrt{s}} = -\frac{1}{2}(\beta_1 u + \beta_2), \quad (19)$$

Integrating from $u(0) = \sqrt{s(0)}$ to $u(T) = 0$ by separating variables,

$$\int_{\sqrt{s(0)}}^0 \frac{2du}{\beta_1 u + \beta_2} = \int_0^T dt, \quad (20)$$

Solving gives

$$T_{NARL+} = \frac{2}{\beta_1} \ln \left(\frac{\beta_1 \sqrt{s(0)} + \beta_2}{\beta_2} \right), \quad (21)$$

When $s < 0$ ($\text{sgn}(s) = -1$), let $v = -s > 0$, then

$$\dot{v} = -\dot{s} = -f(e, -v) - \beta_1 v - \beta_2 \sqrt{v}, \quad (22)$$

Since $\text{sgn}(-v) = -1$ and $f(e, -v) = \frac{\varepsilon_1 |e|}{\varepsilon_2 + (1-k_2)e^{-\delta v}}$, when $v \rightarrow 0$, $f(e, -v) \rightarrow 0$. Thus: $\dot{v} = \beta_1 v + \beta_2 \sqrt{v}$.

Let $w = \sqrt{v}$, then

$$\dot{w} = \frac{1}{2}(\beta_1 w + \beta_2), \quad (23)$$

Integrate from $w(0) = \sqrt{|s(0)|}$ to $w(T) = 0$.

$$\int_{\sqrt{|s(0)|}}^0 \frac{2dw}{\beta_1 w + \beta_2} = -\int_0^T dt, \quad (24)$$

Solving gives

$$T_{NARL-} = \frac{2}{\beta_1} \ln \left(\frac{\beta_1 \sqrt{|s(0)|} + \beta_2}{\beta_2} \right), \quad (25)$$

The proof is completed. $\square$

Combining Equations (3), (14) and (16), we obtain

$$\begin{aligned} \dot{i}_q^* = \frac{J}{1.5p\psi_f} [(c_1 + \alpha\alpha_1|e|^{\alpha_1-1})\dot{e} + c_2e - \frac{B}{J}\omega_m + f(e,s)\operatorname{sgn}(s) + \\ \beta_1s + \beta_2|s|^{0.5}\operatorname{sgn}(s)], \end{aligned} \quad (26)$$

To ensure that the system can reach the sliding surface and converge regardless of the initial state, a Lyapunov function is chosen.

$$V = \frac{1}{2}s^2, \quad (27)$$

Differentiating Equation (25) yields

$$\begin{aligned} \dot{V} &= s\dot{s} \\ &= s(-f(e,s)\operatorname{sgn}(s) - \beta_1s - \beta_2|s|^{0.5}\operatorname{sgn}(s)) \\ &= -f(e,s)|s| - \beta_1s^2 - \beta_2|s|^{1.5}, \end{aligned} \quad (28)$$

Since $f(e,s) > 0$ and $\beta_1, \beta_2 > 0$, for $s \neq 0$, it follows that

$$\dot{V} \leq -\beta_1s^2 - \beta_2|s|^{1.5} < 0, \quad (29)$$

So $\dot{V}$ is strictly negative definite. When $s = 0$, $\dot{V} = 0$, satisfying the equilibrium condition. The overall design block diagram is shown in Figure 2.

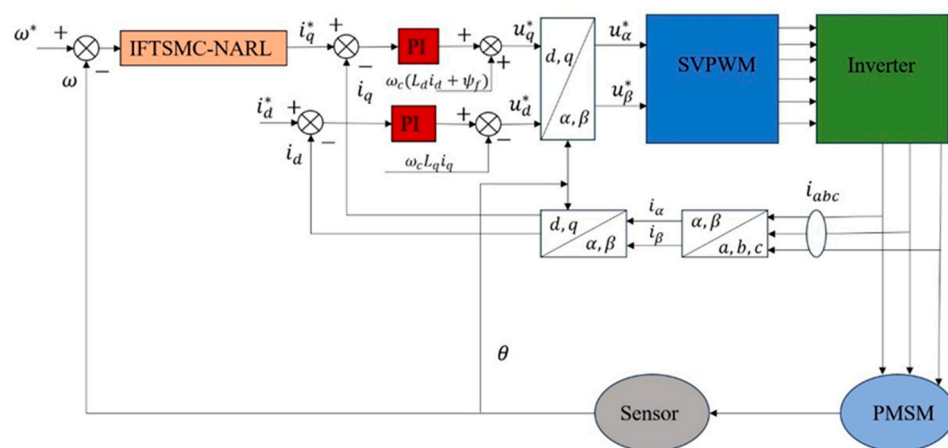


Figure 2. Control Block Diagram of IFSMC-NARL.

4. Experimental Analysis

For comparative purposes, we developed an experimental platform centered on the STM32F4071GT6 microcontroller, as illustrated in Figure 3. The performance of the proposed controller was compared with SMC-TRL, IFSMC-TRL and SMC-[7]. The vector control system employs Space Vector Pulse Width Modulation (SVPWM).

The sliding mode surface and reaching law used in Reference [7] are shown in Formulas (30) and (31).

$$s = \dot{e} + \eta_1 e + \eta_2 \int |e|^{\frac{p}{q}} \operatorname{sgn}(e) dt \quad (30)$$

$$\dot{s} = -\varepsilon \operatorname{sgn}(s) - \lambda s \quad (31)$$

To verify the proposed algorithm, the experiment involves the no-load starting of the motor, with its rotational speed increasing from 0 rpm to 600 rpm. A load torque of 1 N·m is applied at 20 s, and the rotational speed is increased to 1000 rpm at 40 s. The 1 N·m load torque is removed at 60 s, and finally, the rotational speed is reduced to 500 rpm at 80 s. The experimental results are shown in Figures 4 and 5. The parameters of the experimental motor are shown in Table 1, and the parameters involved in different control methods are shown in Table 2.

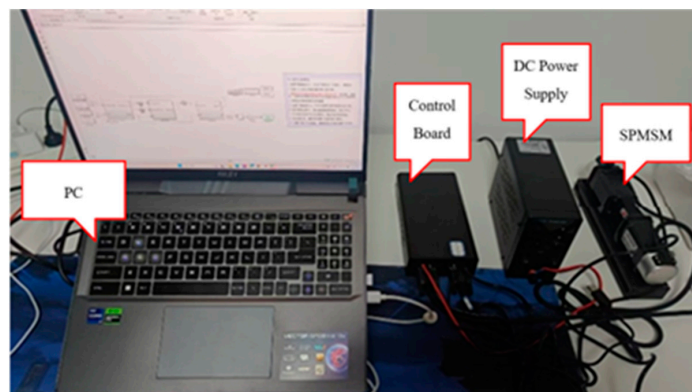


Figure 3. The DSP-based SPMSM experimental platform.

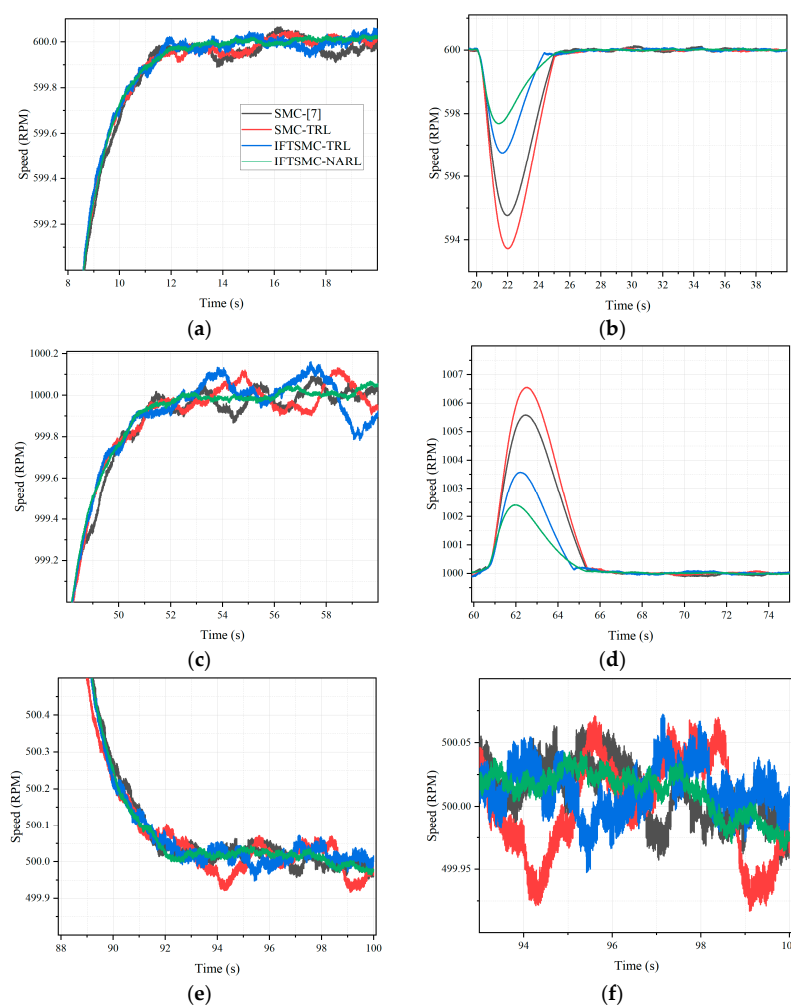


Figure 4. (a) No-load startup; (b) Applying load torque; (c) Speed-up; (d) Removal of load torque; (e) Speed reduction; (f) Steady-state chattering.

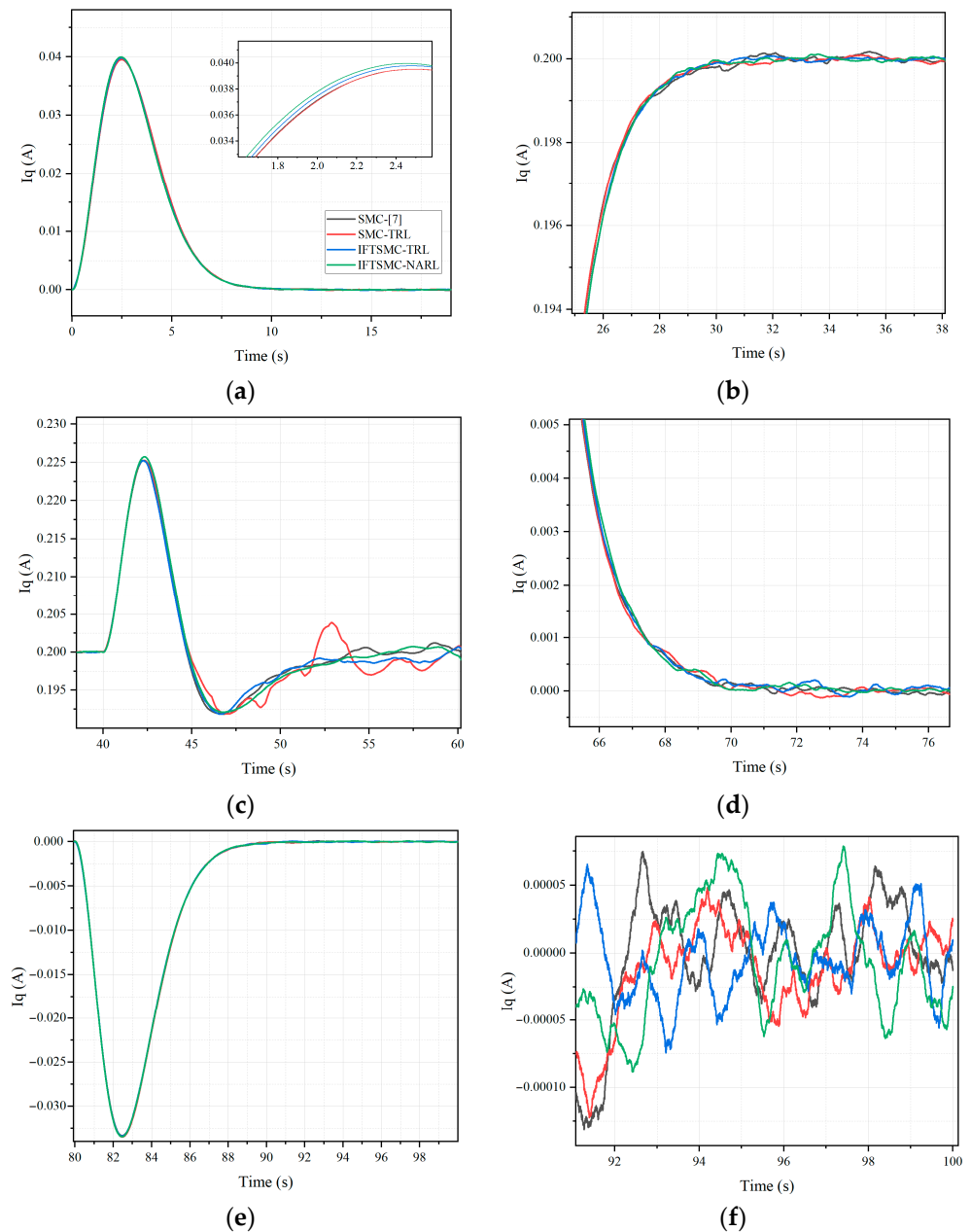


Figure 5. (a) No-load startup current response; (b) Current response under load application; (c) Current response during speed-up; (d) Current response after load removal; (e) Current response during speed reduction; (f) Steady-state current chattering.

Table 1. SPMSM parameters.

Parameters	Values
Number of pole-pairs	4
Stator resistance	0.89 Ω
Stator inductance (d-q axes)	0.62 mH
Permanent magnet flux linkage	0.005 Wb
Inertia	1.7×10^{-4} kg·m ²
Viscous damping	0.000002 N·m·s

Table 2. Controller parameters.

SMC-[7]	$\eta_1 = 12; \eta_2 = 0.01; p = 1; q = 3; k_1 = 3; k_2 = 1$
SMC-TRL	$c_1 = 10; k_1 = 3; k_2 = 1$
IFTSMC-TRL	$c_1 = 12; \alpha_1 = 0.2; \alpha = 0.0001; c_2 = 0.01; k_1 = 3; k_2 = 2$
IFTSMC-NARL	$c_1 = 12; \alpha_1 = 0.2; \alpha = 0.0001; c_2 = 0.01; \varepsilon_1 = 0.05; \varepsilon_2 = 0.2; \delta = 1; \beta_1 = 4; \beta_2 = 1$

In the no-load startup phase, as shown in Figure 4a, the fastest response speed and the smoothest speed rising curve exhibited by IFTSMC-NARL essentially stem from the synergistic effect of its improved fast terminal sliding mode (IFTSM) surface and the novel adaptive reaching law (NARL). The nonlinear term introduced in IFTSM provides strong gain characteristics when the error is large, and in conjunction with the integral feedback mechanism, it accelerates error convergence. Meanwhile, the state-dependent gain adjustment function of NARL promotes the system to quickly approach the sliding mode surface through high-gain switching terms in the large error stage, avoiding the dynamic fluctuations of the traditional SMC-[7] caused by its reliance on fractional-order integral terms, as well as the response delay of SMC-TRL due to the fixed-gain reaching law being unable to match the dynamic changes of errors.

When a 1 N·m load disturbance is applied, as shown in Figure 4b, IFTSMC-NARL has the smallest speed fluctuation and the fastest recovery speed, which benefits from the adaptive adjustment capability of NARL: when a sudden load change causes an increase in error, its gain function dynamically increases with $|e|$, enhancing the disturbance rejection strength, while the nonlinear damping term of IFTSM quickly suppresses error diffusion. Compared with SMC-TRL, which struggles to cope with sudden disturbances due to fixed gains, and IFTSMC-TRL, which exhibits larger fluctuations due to the lack of adaptive gain adjustment, it fully demonstrates the robustness advantage of the proposed algorithm in disturbance suppression.

In the process of increasing the rotational speed to 1000 rpm, as shown in Figure 4c, the high-precision tracking performance of IFTSMC-NARL originates from the continuous elimination of steady-state errors by the integral term in the IFTSM sliding surface, as well as the gain attenuation mechanism of NARL in the error reduction stage. When the system approaches the target speed, $f(e, s)$ weakens high-frequency switching as $|e|$ decreases, avoiding the persistent oscillation of SMC-[7] caused by insufficient design of nonlinear terms and overcoming the defect of insufficient adjustment accuracy of the linear reaching law of SMC-TRL in the small error domain. In the high-speed operation stage after load removal, as shown in Figure 4d, the fluctuation amplitude of only 2 rpm further verifies the strong convergence capability of IFTSM for tiny errors: the gain enhancement effect of the nonlinear term $|e|^{\alpha_1}$ when the error approaches zero, combined with the nonlinear damping term of NARL to suppress high-frequency chattering of the sliding surface, keeps the system stable.

In the process of reducing the rotational speed to 500 rpm, as shown in Figure 4e, the smooth decay characteristic of IFTSMC-NARL benefits from the nonlinear damping adjustment of the $\beta_2 |s|^{0.5}$ term in NARL: when speed changes cause fluctuations in the sliding surface, this damping term strengthens with the increase of $|s|$, suppressing drastic changes. In contrast, traditional methods such as SMC-[7] lead to oscillations due to the lack of such dynamic damping, and SMC-TRL suffers from adjustment lag due to the fixed gain of the reaching law. Finally, in the steady-state operation, as shown in Figure 4f, the minimum chattering phenomenon of IFTSMC-NARL is exactly the comprehensive

result of NARL achieving the dynamic balance of the switching term with “strong gain for large errors and weak gain for small errors” through the state-dependent gain adjustment function, and the integral term of IFTSM eliminating static errors, which effectively reduces the energy loss and the risk of mechanical wear during motor operation.

To more intuitively quantify the performance differences among various control strategies, the test results of key indicators in the above experimental process are summarized in Table 3. This table conducts a quantitative comparison of SMC-[7], SMC-TRL, IFTSMC-TRL, and the proposed IFTSMC-NARL algorithm from three dimensions: response time during no-load startup, speed fluctuation amplitude after load disturbance, and chattering amplitude during steady-state operation. It further verifies the improvement effect of the improved mechanism in the theoretical design on the system’s dynamic performance and robustness. The data shows that the proposed algorithm performs optimally in all indicators, which is consistent with the qualitative analysis conclusions based on experimental phenomena above, fully demonstrating the effectiveness of the collaborative design of the IFTSM sliding mode surface and the NARL reaching law.

Table 3. Control Performance Comparison of Different Control Methods.

Control Method	Response Time (No-Load Startup, s)	Speed Variation After Load Disturbance (rpm)	Amplitude of Chattering
SMC-[7]	12	5.7	0.1
SMC-TRL	16	7.5	0.03
IFTSMC-TRL	12	3.3	0.05
IFTSMC-NARL	11.8	2.3	0.02

Figure 5 illustrates the dynamic response characteristics of the q-axis current i_q under different control strategies, and its variation law is highly consistent with the speed regulation performance. In the no-load startup phase (Figure 5a), the i_q of IFTSMC-NARL can quickly increase to provide sufficient starting torque. Meanwhile, due to the adaptive gain adjustment of NARL, it avoids the current fluctuations of SMC-[7] and the response delay of SMC-TRL, showing fast and smooth characteristics. When the load is applied (Figure 5b), i_q needs to increase in time to compensate for the load torque. By virtue of the nonlinear damping term of IFTSM and the dynamic gain adjustment of NARL, IFTSMC-NARL achieves accurate and small-fluctuation regulation of the current, which is superior to the excessive impact or slow response caused by fixed gains in other methods.

In the processes of speed increase (Figure 5c) and load removal (Figure 5d), the variation of i_q in IFTSMC-NARL always matches the speed target, with smooth transitions and no continuous oscillations. This benefits from the elimination of steady-state errors by the integral term and the strong convergence capability of the nonlinear term for tiny errors. In the speed reduction phase (Figure 5e), the attenuation of its i_q is smooth, avoiding the oscillation phenomenon of SMC-[7]. During steady-state operation (Figure 5f), the chattering amplitude of i_q in IFTSMC-NARL is the smallest, further verifying the suppression effect of NARL on high-frequency switching. It indicates that the proposed algorithm also has significant advantages in current stability control, which echoes the excellent performance of speed regulation.

5. Conclusions

Under complex operating scenarios where PMSM drive systems are subject to internal and external disturbances as well as parameter uncertainties, this paper presents a control

strategy integrating IFTSM and NARL. Via comparative simulation studies against SMC-TRL, SMC-[7], and IFTSMC-TRL, the following specific outcomes are achieved:

1. The multi-modal collaborative design of the improved fast terminal sliding mode surface successfully overcomes the performance bottleneck of traditional terminal sliding mode. This sliding mode surface strengthens the ability to adjust gains in the small error domain through nonlinear terms and offsets steady-state errors via an integral feedback mechanism. It not only retains the finite-time convergence property but also addresses the singularity problem of traditional terminal sliding mode when the error approaches zero. Unlike the sliding mode surface design of SMC-[7], which relies on fractional-order integral terms, the proposed method aligns more accurately with the dynamic change process of errors from large to small by virtue of the collaboration between nonlinear terms and integral terms. It also enhances the convergence capability for tiny errors and improves the system's adaptability to dynamic characteristics under parameter perturbation scenarios.
2. The novel adaptive reaching law attains a dynamic equilibrium between convergence speed and chattering suppression. By means of collaboration between a state-dependent gain adjustment function and a nonlinear damping term, NARL speeds up the convergence to the sliding mode surface with strong gain in the large error stage. It also adaptively reduces gains in the small error stage to weaken high-frequency switching. This overcomes the drawbacks of TRL, where high gain worsens chattering and low gain slows down convergence.
3. The overall control performance is superior under complex operating conditions. Benefiting from the synergistic effect of IFTSM and NARL, the proposed strategy performs well in speed tracking accuracy and chattering attenuation, offering more stable dynamic performance for high-precision servo control.

For future work, on one hand, we will explore the extension of the proposed sliding mode control framework to fixed-time convergence designs, aiming to reduce dependence on initial state information and enhance robustness in dynamic scenarios with abrupt parameter variations, which could further broaden its applicability in PMSM drive systems. On the other hand, we will investigate the integration of the current control strategy with intelligent optimization algorithms (e.g., adaptive particle swarm optimization) to realize self-tuning of key parameters, and validate its performance in multi-motor coordinated control scenarios to meet more complex industrial application requirements.

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